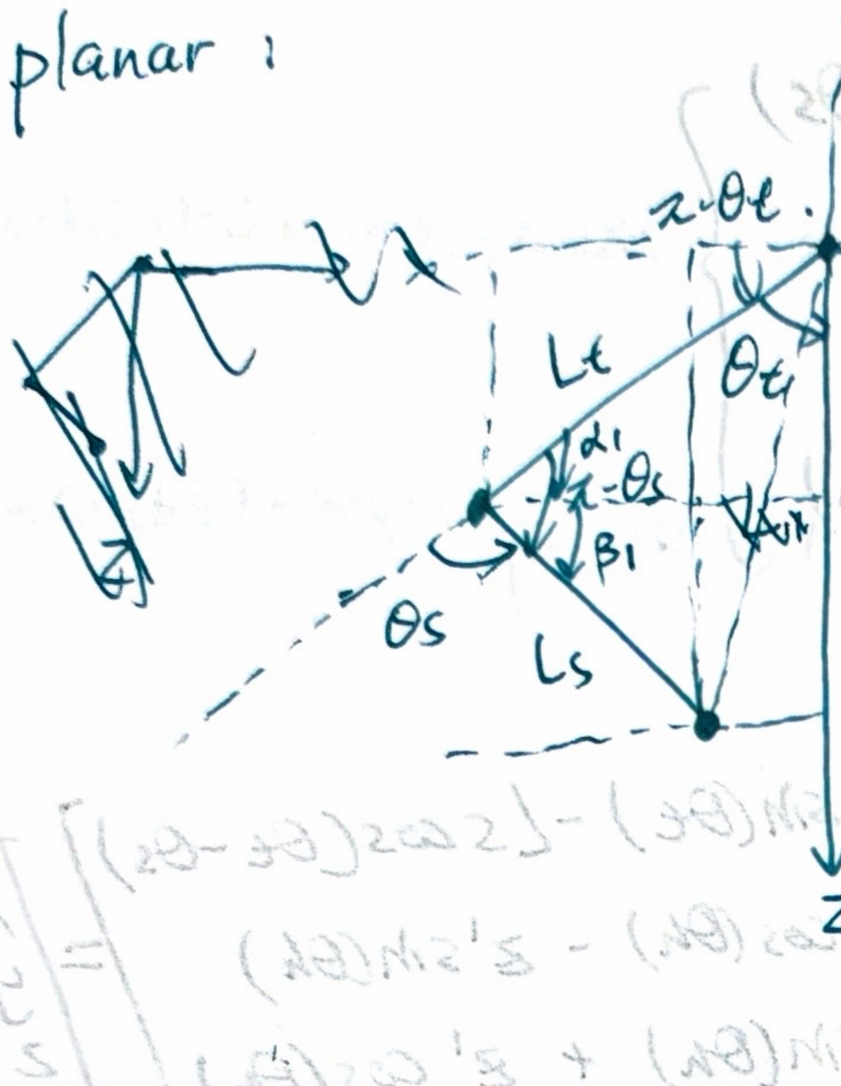
planar :

find x, z ?



$\alpha_1 = \pi - \theta t.$

$\beta_1 = \pi - \theta_s - \alpha_1$

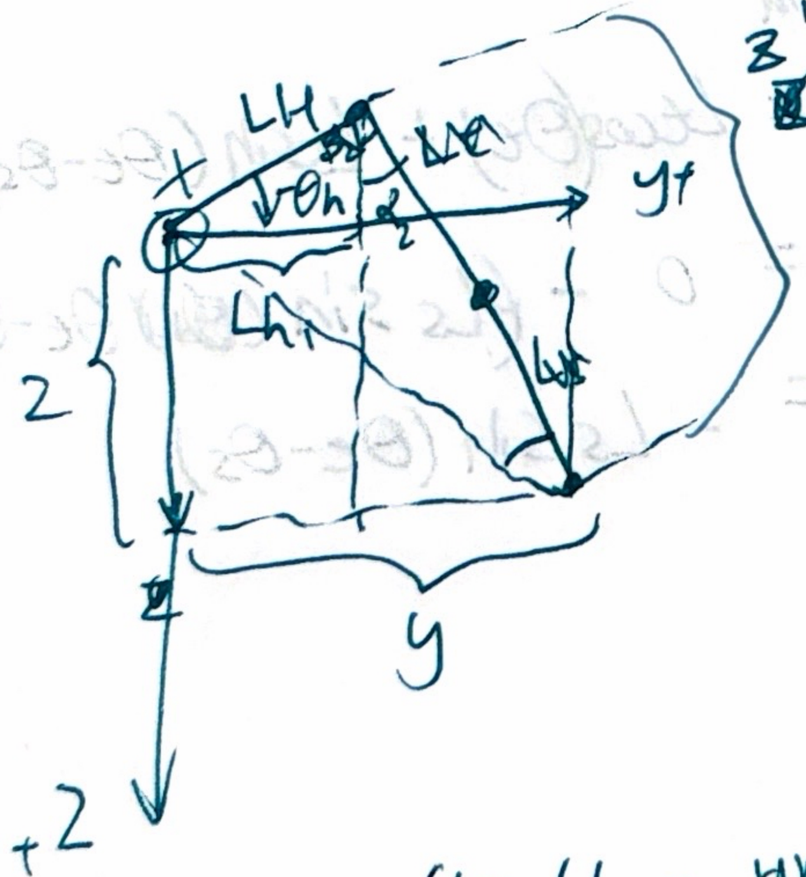
$\beta_1 = \pi - \theta_s - \pi + \theta t = \boxed{\theta t - \theta s.}$

$X t = L t \sin(\theta t).$

$X_s = L s \cdot \cos(\theta t - \theta s).$

So $\begin{cases} X = (X_t - X_s) = L t \sin(\theta t) - L s \cos(\theta t - \theta s). \\ Z' = L t \cos(\theta t) + L s \cdot \sin(\theta t - \theta s). \end{cases}$

Hip :



$Lh_1 = Lh \cdot \cos(-\theta h) = Lh \cos(\theta h).$

$\beta_2 : \beta_2 + |(-\theta h)| = 180° = \pi$

$\begin{cases} |\beta_2| + |\theta h| = \pi/2 \\ |\alpha_2| + |\beta_2| = \pi/2 \end{cases}$

$\theta h - \alpha_2 = 0. \rightarrow |\theta h = \alpha_2|$

$y = Lh_1 + z' \cdot \sin(\theta h) = Lh \cos(\theta h) - z' \sin(\theta h).$

$Z = z' \cos(-\theta h) - LH \sin(-\theta h)$

$Z = z' \cos(\theta h) + LH \sin(\theta h).$

D Fk:

$$\left.\begin{aligned} x &= L_t \sin(\theta_t) - L_s \cos(\theta_t - \theta_s) \\ y &= L_h \cos(\theta_h) - z' \sin(\theta_h) \\ z &= z' \cos(\theta_h) + L_4 \sin(\theta_h) + z' \cos(\theta_h) \end{aligned}\right\}$$

so Fk position transformation A:

$$\vec{\theta} = \begin{bmatrix} \theta_h \\ \theta_t \\ \theta_s \end{bmatrix} \qquad \vec{P}(\theta_h, \theta_s, \theta_t) = \begin{bmatrix} L_t \sin(\theta_t) - L_s \cos(\theta_t - \theta_s) \\ L_h \cos(\theta_h) - z' \sin(\theta_h) \\ L_h \sin(\theta_h) + z' \cos(\theta_h) \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$z' = L_t \cos(\theta_t) + L_t L_s \sin(\theta_t - \theta_s)$$

differentiate to get Jacobian J:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_h} & \frac{\partial x}{\partial \theta_t} & \frac{\partial x}{\partial \theta_s} \\ \frac{\partial y}{\partial \theta_h} & \frac{\partial y}{\partial \theta_t} & \frac{\partial y}{\partial \theta_s} \\ \frac{\partial z}{\partial \theta_h} & \frac{\partial z}{\partial \theta_t} & \frac{\partial z}{\partial \theta_s} \end{bmatrix} \Rightarrow \begin{cases} \frac{\partial x}{\partial \theta_h} = 0 \\ \frac{\partial x}{\partial \theta_t} = L_t \cos(\theta_t) + L_s \sin(\theta_t - \theta_s) \\ \frac{\partial x}{\partial \theta_s} = 0 - (L_s \sin(\theta_t - \theta_s)) \\ \frac{\partial x}{\partial \theta_s} = -L_s \sin(\theta_t - \theta_s) \end{cases}$$

$$\frac{\partial y}{\partial \theta_h} = -L_h \sin(\theta_h)$$

$$y = L_h \cos(\theta_h) - \left(L_t \cos(\theta_t) \cdot \sin(\theta_h) + L_s \sin(\theta_t - \theta_s) \cdot \sin(\theta_h) \right)$$

$$\frac{\partial y}{\partial \theta_h} = -L_h \sin(\theta_h) - \left(L_t \cos(\theta_t) \cos(\theta_h) + L_s \sin(\theta_t - \theta_s) \cos(\theta_h) \right)$$

$$\frac{\partial y}{\partial \theta_h} = -L_h \sin(\theta_h) - \cos(\theta_h) \cdot \left(L_t \cos(\theta_t) + L_s \sin(\theta_t - \theta_s) \right)$$

$$\frac{\partial y}{\partial \theta_t} = + L_t \sin(\theta_t) \cdot \sin(\theta_h) - L_s \cos(\theta_t - \theta_s) \cdot \sin(\theta_h)$$

$$\frac{\partial y}{\partial \theta_s} = -\left(L_s \cdot \cos(\theta_t - \theta_s) \cdot \sin(\theta_h)(-1) \right) = L_s \cos(\theta_t - \theta_s) \cdot \sin(\theta_h)$$

$$z = L_h \sin(\theta_h) + L_t \cos(\theta_t) \cdot \cos(\theta_h) + L_s \sin(\theta_t - \theta_s) \cdot \cos(\theta_h)$$

$$\frac{\partial z}{\partial \theta_h} = L_h \cos(\theta_h) - L_t \cos(\theta_t) \cdot \sin(\theta_h) - L_s \sin(\theta_t - \theta_s) \sin(\theta_h)$$

$$\frac{\partial z}{\partial \theta_t} = -L_t \sin(\theta_t) \cdot \cos(\theta_h) + L_s \cos(\theta_t - \theta_s) \cdot \cos(\theta_h)$$

$$\frac{\partial z}{\partial \theta_s} = L_s \sin - L_s \cos(\theta_t - \theta_s) \cos(\theta_h)$$

$$J = \begin{bmatrix} 0 & L_t\cos(\theta_t) + L\sin(\theta_t - \theta_s) & -L_s\sin(\theta_t - \theta_s) \\ -L_h\sin(\theta_h) - \cos(\theta_h)\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s)\right) & L_t\sin(\theta_t)\sin(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t - \theta_s) & L_s\sin(\theta_h)\cos(\theta_t - \theta_s) \\ L_h\cos(\theta_h) - \sin(\theta_h)\cdot\left(L_t\cos(\theta_t) + L\sin(\theta_t - \theta_s)\right) & L_s\sin(\theta_h)\cos(\theta_t - \theta_s) & -L_s\cos(\theta_h)\cos(\theta_t - \theta_s) \end{bmatrix}$$

$$J = \begin{bmatrix} 0 & L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s) & -L_s\sin(\theta_t - \theta_s) \\ -L_h\sin(\theta_h) - \cos(\theta_h)\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s)\right) & L_t\sin(\theta_t)\sin(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t - \theta_s) & L_s\sin(\theta_h)\cos(\theta_t - \theta_s) \\ L_h\cos(\theta_h) - \sin(\theta_h)\cdot\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s)\right) & L_t\sin(\theta_h)\cos(\theta_t - \theta_s) & -L_s\cos(\theta_h)\cos(\theta_t) \end{bmatrix}$$